

1. Categories of Errors in Java

Java errors are generally divided into **three main types**:

Error Type	When It Occurs	Detected By	Examples
Compile-time Errors	During compilation (before running)	Compiler	Syntax mistakes, missing semicolons, undeclared variables
Run-time Errors	While the program is running	JVM	Division by zero, null pointer access, array index out of bounds
Logical Errors	Program runs but produces wrong output	Programmer	Wrong formula, incorrect loop condition

2. Compile-time Errors

These are **syntax or semantic mistakes** detected by the **Java compiler (javac)**.

The program will **not execute** until these errors are fixed.

Common Causes:

- **Syntax errors**
- **Type mismatches**
- **Accessing undeclared variables**
- **Wrong method signatures**
- **Missing return statements in non-void methods**

Examples:

```
// Missing semicolon
```

```
int x = 5 // ERROR: ';' expected
```

```
// Undeclared variable
```

```
System.out.println(num); // ERROR: cannot find symbol 'num'
```

```
// Type mismatch
```

```
int n = "Hello"; // ERROR: incompatible types
```

```
// Missing return
```

```
int square(int n) {  
    n * n; // ERROR: missing return statement  
}
```

3. Run-time Errors

These occur **after successful compilation**, when the **JVM executes** the program. They are usually caused by **invalid operations** or **bad input data**.

Examples:

// Division by zero

```
int a = 10 / 0; // java.lang.ArithmeticException
```

// Null Pointer

```
String s = null;
```

```
System.out.println(s.length()); // java.lang.NullPointerException
```

// Array index out of bounds

```
int arr[] = {1, 2, 3};
```

```
System.out.println(arr[5]); // java.lang.ArrayIndexOutOfBoundsException
```

// Number format

```
String num = "abc";
```

```
int x = Integer.parseInt(num); // java.lang.NumberFormatException
```

4. Logical Errors

- **No compiler or runtime error**
- Program executes but **output is wrong**
- Often caused by **wrong algorithm, wrong condition, or calculation mistakes**
- The hardest to find — requires **debugging and testing**

Example:

// Wrong logic: intended to find square

```
int square(int n) {  
    return n + n; // Wrong, should be n * n  
}
```

If you pass 5, output will be 10 instead of 25.

Find Errors:

- 1.**
`int arr = {1, 2, 3, 4};`
`System.out.println(arr[0]);`
- 2.**
`String s = "HELLO";`
`System.out.println(s.charAt(10));`
- 3.**
`String s = "WORLD";`
`System.out.println(s.substring(3, 10));`
- 4.**
`String name = null;`
`System.out.println(name.length());`
- 5.**
`int[] nums = {10, 20, 30};`
`System.out.println(nums[5]);`
- 6.**
`int a = 5 / 0;`
`System.out.println(a);`
- 7.**
`int[] arr;`
`arr[0] = 10;`
`System.out.println(arr[0]);`
- 8.**
`String str = "Java";`
`str.toLowerCase();`
`System.out.println(str);`
- 9.**
`int arr[] = new int[3];`
`for(int i = 0; i <= 3; i++) {`
 `arr[i] = i + 1;`
`}`
- 10.** `String s = "Programming";`
`System.out.println(s.substring(-2, 5));`
- 11.** `String text = "abc";`
`int num = Integer.parseInt(text);`
`System.out.println(num);`
- 12.**
`String s = "abcdef";`
`System.out.println(s.charAt(-1));`
- 13.**
`int arr[] = {1, 2, 3};`
`System.out.println(arr[-1]);`
- 14.**
`String s = "JAVA";`
`System.out.println(s.substring(2, 1));`
- 15.**
`int a = "50";`
`System.out.println(a);`
- 16.**
`String s = "abcd";`
`System.out.println(s.length);`
- 17.** `String s = "abcd";`
`System.out.println(s.charAt(4));`
- 18.**
`int arr[] = {1, 2, 3};`
`System.out.println(arr.length());`
- 19.**
`String s = "abcdef";`
`for(int i = 0; i <= s.length(); i++) {`
 `System.out.println(s.charAt(i));`
`}`
- 20.**
`String s = "Hello";`
`if(s == "Hello") {`
 `System.out.println("Match");`
`} else {`
 `System.out.println("No match");`
`}`
- 21.** `int a=10, b=5;`
 `int k= (a+b)/p;`
 `int p= a-b;`
- 22.** `int a=a+10;`
- 23.** `int a[]=new int[5];`
 `a[]={1,2,3,4,5};`
 `System.out.println(a[3]);`
- 24.** `int a=Math.cbrt(125);`

Other Snippets:

```
1. System.out.println(Math.pow(36, 1/2)+4);

2. System.out.println(Math.pow(36, 1.0/2)+4);

3. System.out.println(Math.sqrt(-49));

4. System.out.println(Math.random()*5+4);
   Write the possible range of values

5. int k = (2 < 4) ? ((2 < 3) ? 2 : ((3 < 5) ? 4 : 0)) : 0;
   k=?

6. int x=10, y=4;
   y -= x++ + ++x;
   z=?

7. int x= 5/2 + 2/5 + 5%2 + 2%5;

   x=?

8. System.out.println(1+2+ "JAVA" + (1+2)+3);

9. Write using JAVA  $\frac{\sqrt[4]{x}}{|y^z|}$ 

10. for(int i = 1; i <= 3; i++) {
    for(int j = 1; j <= 3; j++) {
        System.out.print((i * j) + " ");
    }
    System.out.println();
}

11. for(int i = 1; i <= 3; i++) {
    for(int j = 1; j <= 3; j++) {
        System.out.print(i + "" + j + " ");
    }
}

12. int a=10, b=10;

    for(int i=1; i<=5; i++)

        a+=2;

        b-=2;

System.out.println(a+b);
```

13.

```
class Sample {
    int value;

    Sample() {
        value = 5;

        value += 3;

        System.out.println("Default Constructor: " +
value);
    }

    Sample(int x) {
        value = x;

        value *= 2;

        System.out.println("Constructor: " + value);
    }

    public static void main(String[] args) {
        Sample obj1 = new Sample();

        Sample obj2 = new Sample(4);
    }
}
```

14.

```
class DoWhile {
    public static void main(String[] args) {
        int x = 5;

        int count = 0;

        do {
            count += x;

            x--;
        } while(x > 0);

        System.out.println(count);
    }
}
```